

Software Engineer – Interview Prep Guide

Q: What is the role of a software engineer?

A: They design, develop, test, and maintain software applications that solve business or user problems.

Q: What languages are you proficient in?

A: C++, Java, Python, JavaScript, and frameworks like React, Django, or Spring depending on project needs.

Q: Explain the software development lifecycle (SDLC).

A: Phases include Planning → Analysis → Design → Development → Testing → Deployment → Maintenance.

Q: What is Agile methodology?

A: An iterative development approach where requirements and solutions evolve through collaboration and sprints.

Q: How do you ensure code quality?

A: By writing unit tests, conducting code reviews, using linters, and following clean coding practices.

Q: What's the difference between functional and non-functional requirements?

A: Functional = what the system should do, Non-functional = performance, security, scalability, etc.

Q: How do you debug complex issues?

A: By reproducing errors, analyzing logs, using debuggers, and isolating problem modules.

Q: Describe a challenging software project you worked on.

A: I built a scalable microservices system for an e-commerce client, handling high traffic while maintaining low latency.